

You already built this toolkit in Week 3 (TMWR Ch. 9–10). This page is a working reference, not new material — use it to refit the same metrics and resampling workflow to today's model.

MetricsTMWR Ch. 9

Regression

```
metric_set(rmse, rsq, mae)
reg_preds |> reg_metrics(truth = y, estimate = .pred)
```

METRICREADS AS

RMSE

typical error, in original units; punishes big misses hardest

R²

% of variance explained; unitless, 0–1

MAE

typical error, robust to outliers

Binary Classification

```
conf_mat(preds, truth = y, estimate = .pred_class)
metric_set(accuracy, sens, spec, precision)
```

METRICMARKETING QUESTION

Sensitivity

of true responders, how many did we catch?

Specificity

of true non-responders, how many did we skip?

Precision

of who we targeted, how many responded?

Accuracy

overall correct — caution, see below

ACCURACY TRAP

With a 3–8% response rate, predicting "no" for everyone scores 92–97% accuracy and is useless. Always pair accuracy with sensitivity/specificity.

ROC / AUC – Threshold-Free View

```
roc_curve(preds, truth = y, .pred_yes)
roc_auc(preds, truth = y, .pred_yes)
```

AUCREAD

0.50

coin flip

> 0.65

usable for targeting

> 0.75

quite good

1.00

perfect (rare)

Multiclass

Same metrics, computed one-vs-rest per class, then **macro-averaged** by default — every class counts equally, even rare ones (e.g. Platinum tier).

ResamplingTMWR Ch. 10

WHY BOTHER

A single test-set number is one draw from a distribution. Resampling gives you the *mean and spread* of likely performance — and never touches the test set, which is reserved for one final check.

Build Folds

```
vfold_cv(train, v = 10, strata = y)
```

Fit + Score Across Folds

```
fit_resamples(
  wf, resamples = folds,
  metrics = metric_set(accuracy, roc_auc),
  control = control_resamples(save_pred = TRUE)
)
collect_metrics(cv_results)
```

RESUBSTITUTION WARNING

Never trust metrics computed on the same data the model was trained on — they're optimistic. Always score on assessment folds or the held-out test set.

Quick Decision Guide

Outcome is continuous (spending, revenue)?

→ metric_set(rmse, rsq, mae)

Outcome is yes/no (response, churn, purchase)?

→ metric_set(accuracy, sens, spec, roc_auc)

Outcome has 3+ categories (tier, segment)?

→ same functions; check macro vs. micro averaging

Reporting a single trustworthy number?

→ fit_resamples() + collect_metrics(), not resubstitution

Comparing several models or tuning hyperparameters?

→ resample every candidate the same way; save the test set for the very end

IBM 6540 · Applied Machine Learning for Marketing

Full treatment: TMWR Ch. 9–10 (Week 3 notebook)